USN | | | | | | | | | | [20EEDE741]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2023

OPERATIONS RESEARCH

Time: 3 Hours **Maximum Marks: 70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.
- iii. Normal distribution table will be provided.

1. a. Explain some of the applications of operations research to engineering and managerial problems **08**

b. Solve the following linear programming problem graphically: **06**

Maximise $Z = 4x + y$ ... (1)

subject to the constraints:

$x + y \leq 50$ ... (2)

$3x + y \leq 90$ ... (3)

$x \geq 0, y \geq 0$ ... (4)

2. a. A dairy firm has three plants located in a state. The daily milk production at each plant is as follows: **09**

Plant 1 : 6 million litres, Plant 2 : 1 million litres, and Plant 3 : 10 million litres

Each day, the firm must fulfil the needs of its four distribution centres.

The minimum requirement of each centre is as follows:

Distribution centre 1 : 7 million litres, Distribution centre 2 : 5 million litres, Distribution centre 3 : 3 million litres, and Distribution centre 4 : 2 million litres

Cost (in hundreds of rupees) of shipping one million litre from each plant to each distribution centre is given in the following table:

		Distribution Centre			
		D_1	D_2	D_3	D_4
Plant	P_1	2	3	11	7
	P_2	1	0	6	1
	P_3	5	8	15	9

Find the initial basic feasible solution for given problem by using following methods:

- **i.** North-west corner rule
- **ii.** Least cost method

iii. Vogel's approximation method

b. Define transportation model, represent it in the form of general matrix **05**

3. a. Explain the steps followed in Hungarian method **06**

b. Solve the following assignment problem. Cell values represent cost of assigning jobs A, B, C and D to the machines I, II, III and IV. **08**

machines

jobs	I	II	III	IV
A	10	12	19	11
B	5	10	7	8
C	12	14	13	11
D	8	15	11	9

OR

4. a. Differentiate between transportation model and assignment model **06**

b. Solve the travelling salesman problem given by the following data. **08**

C12 = 20, C13 = 4, C14 = 10, C23 = 5, C34 = 6, C25 = 10, C35 = 6, C45 = 20 where Cij = Cji and there is no route between cities I and j if the value for Cij is not shown.

5. a. A Project is composed of seven activities whose time estimates are listed in the following table. Activities are simplified by this beginning (1) ones ending (j) Node member. Calculate expected project length. **08**

Activity		Estimated duration in weeks		
i	j	Optimistic	Most likely	Pessimestic
1	2	1	1	7
1	3	1	4	7
1	4	2	2	8
2	5	1	1	1
3	5	2	5	14
4	6	2	5	8
5	6	3	6	15

b. Define a project. Explain the basic steps in project management. **06**

OR

6. a. Explain the steps involved in Fulkerson's rule (i - j) rule. **04**

b. A project has the following times schedule **10**

Activity	Times in weeks	Activity	Times in weeks
(1 – 2)	4	(5 – 7)	8
(1 – 3)	1	(6 – 8)	1
(2 – 4)	1	(7 – 8)	2
(3 – 4)	1	(8 – 9)	1
(3 – 5)	6	(8 – 10)	8
(4 – 9)	5	(9 – 10)	7
(5 – 6)	4		

Construct the network and compute

i. TE and TL for each event

ii. Float for each activity

iii. Critical path and its duration

7. a. Explain the maximin – minimax principle of game theory. How is the game theory problems classified? **05**

b. Find Solution of game theory problem using dominance and graphical method **09**

Player A\Player B	B1	B2	B3	B4
A1	2	2	3	-1
A2	4	3	2	6

OR

8. a. Two players A and B put a coin each simultaneously on the table without showing each other. A wins Rs.8/- when both coins show head and Rs.1/- when both coins are tail. Player B wins Rs.3/- when the coins do not match. Prepare a payoff matrix and determine the best strategy of each player and value of the game. **05**

Player A\Player B	B1	B2
A1	8	-3
A2	-3	1
A3	8	1

b. Use the graphical method to reduce the following game and hence solve. **09**

$$\begin{bmatrix} 3 & -3 & 4 \\ -1 & 1 & -3 \end{bmatrix}$$